14. Depth First Search

Problem Statement

Write Recursive Approach for Depth First Search (DFS).

Given a graph, perform Depth First Search (DFS) traversal on the graph using a recursive approach.

Example 1:

Graph:



Output: 1 2 5 4 3

Explanation: Starting from node 1, we visit its adjacent nodes in order: 2, 5, and 4. From node 4, we visit node 3.

Example 2:

Graph:



Output: 0 1 3 2

Explanation: Starting from node 0, we visit its adjacent nodes in order: 1 and 2. From node 1, we visit node 3.

Example 3:

Graph:



Output: 0 1 2 3

Explanation: Starting from node 0, we visit its adjacent nodes in order: 1 and 3. From node 1, we visit node 2.

Try it yourself

Try solving this question here:

Java

```java
import java.util.ArrayList;
import java.util.HashSet;
import java.util.List;
import java.util.Set;

class Solution {
    public List<Integer> DFS(List<List<Integer>> args, int n, int first) {
        List<Integer> depthFirstSearch = new ArrayList<>();
        // TODO: Write your code here
        return depthFirstSearch;
    }
}
```

Test Case | Run Code Result | Stdout

```
[[1,2],[1,5],[2,5],[1,4],[4,3],[3,5]]
6
1
```

Use Example Testcases | Run | Submit